



Laws of Exponents

Algebra Worksheet · Grade 8-10

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Learning Objectives

- Apply the seven laws of exponents to simplify algebraic expressions
- Distinguish between adding and multiplying exponents in different contexts
- Evaluate expressions involving zero, negative, and positive exponents

Simplify each expression using the laws of exponents and write your answer in simplest form.

1. Simplify using the zero exponent rule.

$$(5x)^0$$

Answer: _____

2. Simplify using the product rule of exponents.

$$2x \cdot x$$

Answer: _____

3. Simplify using the product rule of exponents.

$$m^3 \cdot m$$

Answer: _____

4. Simplify using the product rule of exponents.

$$x^2 \cdot x^5$$

Answer: _____

5. Simplify using the quotient rule of exponents.

$$\frac{x^6}{x^2}$$

Answer: _____

6. Simplify using the power of a power rule.

$$(x^2)^5$$

Answer: _____

7. Simplify using the power of a product rule.

$$(5x)^2$$

Answer: _____



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8. Rewrite using the negative exponent rule.

$$2^{-1}$$

Answer: _____

9. Evaluate using the rule for exponents of one.

$$7^1$$

Answer: _____

10. Simplify by combining the power of a product and power of a power rules.

$$(2x^3)^4$$

Answer: _____



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Remind students to distinguish between the product rule (add exponents) and the power-of-a-power rule (multiply exponents).

Solutions

1. Simplify using the zero exponent rule.

$$(5x)^0$$

- Any nonzero expression raised to the zero power equals one.
- Therefore the entire expression simplifies to 1.

Answer: 1

2. Simplify using the product rule of exponents.

$$2x \cdot x$$

- Both x terms have an exponent of one.
- Add the exponents: one plus one equals two.
- The coefficient 2 remains, giving $2x$ squared.

Answer: $2x^2$

3. Simplify using the product rule of exponents.

$$m^3 \cdot m$$

- The variable m has an implied exponent of one.
- Add the exponents: three plus one equals four.
- The simplified expression is m to the fourth power.

Answer: m^4

4. Simplify using the product rule of exponents.

$$x^2 \cdot x^5$$

- Since the bases are the same, add the exponents.
- Two plus five equals seven.
- The simplified expression is x to the seventh power.

Answer: x^7

5. Simplify using the quotient rule of exponents.

$$\frac{x^6}{x^2}$$

- Subtract the denominator exponent from the numerator exponent.
- Six minus two equals four.
- The simplified expression is x to the fourth power.

Answer: x^4



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6. Simplify using the power of a power rule.

$$(x^2)^5$$

- When a power is raised to another power, multiply the exponents.
- Two times five equals ten.
- The simplified expression is x to the tenth power.

Answer: x^{10}

7. Simplify using the power of a product rule.

$$(5x)^2$$

- Distribute the exponent of two to both the 5 and the x .
- Five squared is twenty-five and x squared remains.
- The simplified expression is $25x$ squared.

Answer: $25x^2$

8. Rewrite using the negative exponent rule.

$$2^{-1}$$

- A negative exponent means take the reciprocal of the base.
- Two to the negative one becomes one over two.
- The simplified value is one half.

Answer: $\frac{1}{2}$

9. Evaluate using the rule for exponents of one.

$$7^1$$

- Any number raised to the first power equals itself.
- Therefore seven to the first power is seven.

Answer: 7

10. Simplify by combining the power of a product and power of a power rules.

$$(2x^3)^4$$

- Distribute the exponent four to both the 2 and the x cubed.
- Two to the fourth power equals sixteen.
- Multiply the exponents three and four to get x to the twelfth power.
- The simplified expression is sixteen x to the twelfth.

Answer: $16x^{12}$



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